BloodBridge : AI-Enabled Blood Bank Management System

1. Project Description

BloodBridge is a web-based Blood Bank Management System that connects blood donors, patients, hospitals, and blood banks through a centralized platform. The system enables users to register as donors, request blood during emergencies, and manage blood availability in real time.

The platform reduces delays during medical emergencies by identifying available donors quickly and notifying them about urgent blood requirements.

The system is built using **Python Flask, HTML, CSS, JavaScript, and cloud services like AWS EC2 and DynamoDB**. It provides a scalable and efficient way to manage blood donation and request processes digitally.

Key Features

- Donor Registration and Login
- Blood Request Management
- Blood Inventory Monitoring
- Donor Search based on Blood Group
- Emergency Notification System
- Admin Dashboard for Management

2. System Architecture The BloodBridge system follows a three-layer architecture:
3. User Interface Layer Technologies used:

HTML

CSS

JavaScript

Responsibilities:

User registration and login

Blood request form

Dashboard display

2. **Application Layer** (Backend) Technologies used:

Python

Flask Framework

Responsibilities:

Handle HTTP requests

Process blood requests

Communicate with database

Manage authentication

3. **Database Layer Technologies used:**

AWS DynamoDB / MySQL

Responsibilities:

Store donor details

Store blood request data

Maintain blood inventory

Manage user records

3. Tools and Technologies Used

Category	Technology
Programming Language	Python
Backend Framework	Flask
Frontend	HTML, CSS, JavaScript
Database	AWS DynamoDB / MySQL
Cloud Deployment	AWS EC2
Development Tool	VS Code
Version Control	GitHub

4. **Project Workflow**

Phase 1: Environment Setup Install Python

Install Flask and dependencies

Setup project folder

Phase 2: Backend Development Implement user authentication

Create donor registration module

Implement blood request module

Connect database

Phase 3: Frontend Development Create login and registration pages

Create request blood interface

Develop dashboard

Phase 4: Deployment Launch AWS EC2 instance

Deploy Flask application

Connect to database

Phase 5: Testing Test user registration

Test blood request submission

Test donor search functionality

```
bloodbank/
│
├── app.py
├── database.py
├── templates/
│      ├── login.html
│      ├── register.html
│      ├── dashboard.html
│
├── static/
│      ├── style.css
│      ├── script.js
│
└── requirements.txt
```

6. Backend Implementation

6.1 Flask Application Initialization

```python
import boto3
from flask import Flask, render_template, request, redirect, url_for, flash, session
from datetime import datetime
import uuid
from boto3.dynamodb.conditions import Attr

app = Flask(__name__)
app.secret_key = "bloodbridge_secret"

# ---------------- DynamoDB Setup ----------------

aws_region = "ap-south-1"

dynamodb = boto3.resource(
    "dynamodb",
    region_name=aws_region
)

users_table = dynamodb.Table("users")
requests_table = dynamodb.Table("requests")
```

6.2 User Registration Module

This module allows donors to register with their personal information and blood group.

```
@app.route('/register', methods=['POST'])
def register():
    name = request.form['name']
    blood_group = request.form['blood_group']
    phone = request.form['phone']

    conn = sqlite3.connect("database.db")
    cur = conn.cursor()

    cur.execute("INSERT INTO donors(name, blood_group, phone) VALUES(?
                (name, blood_group, phone))

    conn.commit()
    conn.close()
```

6.3 Blood Request Module

Patients can request blood through this module.

```
@app.route('/request_blood', methods=['POST'])
def request_blood():
    patient_name = request.form['patient_name']
    blood_group = request.form['blood_group']
    location = request.form['location']

    conn = sqlite3.connect("database.db")
    cur = conn.cursor()

    cur.execute("INSERT INTO requests(patient_name,blood_group,location) VALUES(
                (patient_name, blood_group, location))

    conn.commit()
    conn.close()
```

7. Frontend Implementation

The frontend provides user interaction with the system.

Login Page

```
<form method="post" action="/login">
  <input type="text" name="email" placeholder="Email">
  <input type="password" name="password" placeholder="Password">
  <button type="submit">Login</button>
</form>
```

8. CSS Styling

CSS improves the visual appearance of the application.

```
body{
  background-color:#f4f4f4;
  font-family:Arial;
}

form{
  width:300px;
  margin:auto;
  padding:20px;
  background:white;
}
```

9.Deployment

The application is deployed using AWS EC2.

Steps

1 Launch EC2 Instance

2.Configure Security Group (Allow port 5000)

3.Upload project files

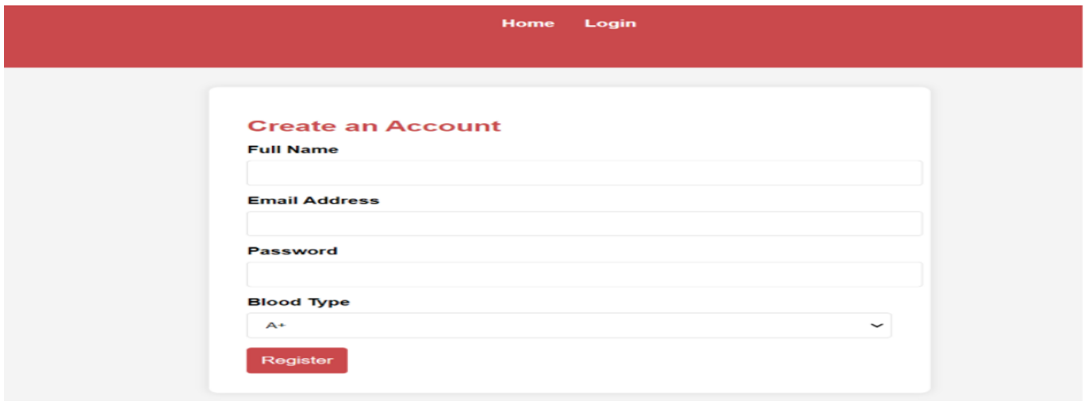
4.Install dependencies

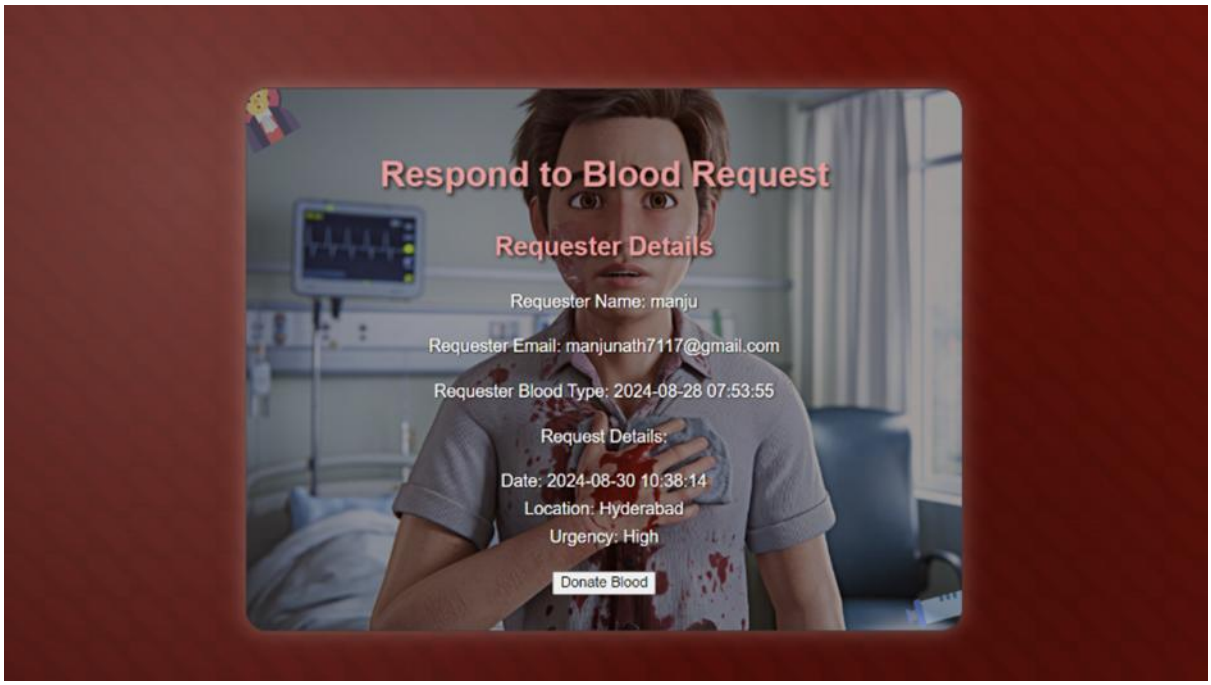
```
pip install flask
python app.py
```

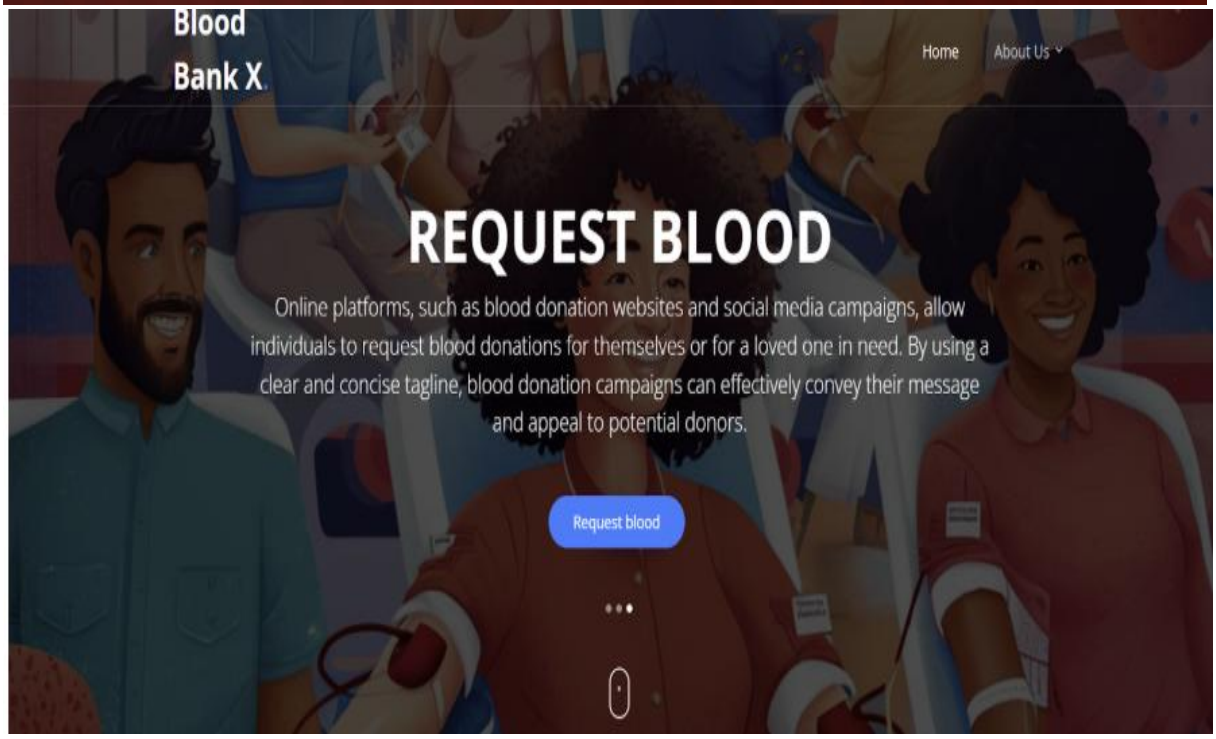
10 Testing

Testing ensures that the system works correctly.

```
home/ec2-user/.local/lib/python3.9/site-packages/boto3/compat.py:8
ing April 29, 2026. To continue receiving service updates, bug fix
rmation can be found here: https://aws.amazon.com/blogs/developer/
 warnings.warn(warning, PythonDeprecationWarning)
* Serving Flask app 'app'
* Debug mode: on
ARNING: This is a development server. Do not use it in a productio
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.32.74:5000
ress CTRL+C to quit
* Restarting with stat
home/ec2-user/.local/lib/python3.9/site-packages/boto3/compat.py:8
ing April 29, 2026. To continue receiving service updates, bug fix
rmation can be found here: https://aws.amazon.com/blogs/developer/
 warnings.warn(warning, PythonDeprecationWarning)
* Debugger is active!
* Debugger PIN: 106-561-583
```


Home
Login
Create an Account
Full Name
Email Address
Password
Blood Type
A+
Register


Respond to Blood Request
Requester Details
Requester Name: manju
Requester Email: manjunath7117@gmail.com
Requester Blood Type: 2024-08-28 07:53:55
Request Details:
Date: 2024-08-30 10:38:14
Location: Hyderabad
Urgency: High
Donate Blood


Blood
Bank X
Home
About Us
REQUEST BLOOD
Online platforms, such as blood donation websites and social media campaigns, allow individuals to request blood donations for themselves or for a loved one in need. By using a clear and concise tagline, blood donation campaigns can effectively convey their message and appeal to potential donors.
Request blood

Conclusion

BloodBridge provides a reliable platform for managing blood donations and requests efficiently. The system reduces the time required to locate donors and improves communication between blood banks, hospitals, and donors.

By integrating web technologies with cloud infrastructure, the platform offers a scalable solution that can support emergency healthcare services.

Future development can further enhance the system with intelligent donor matching and mobile accessibility.